Youkai Zhou

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EDUCATION

Drexel University Philadelphia, PA
Master of Science in Computer Science *Expected June 2028*

University of Mississippi Oxford, MS
Bachelor of Science in Computer Science | GPA: 3.5/4.0 *May 2026*

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Machine Learning: LightGBM, scikit-learn, pandas, NumPy, Walk-Forward Validation
Web Development: React, Node.js, FastAPI, Fastify
Data Storage: PostgreSQL, DuckDB, Parquet, Redis, SQLite
Developer Tools: Git, Linux, Docker, AWS EC2
Languages: English (Fluent), Mandarin (Native)

WORK EXPERIENCE

Ichiban Asian Bistro and Go Madison, MS | May 2022 - August 2026
Assistant Manager (Seasonal) *December 2024 - August 2026*
Host and Front Desk Cashier (Seasonal) *May 2022 - December 2024*

- Built an Excel reservation tracker for front-desk bookings, speeding up seating for 150+ daily customers
- Compiled kitchen and bar inventory counts into stock reports, supporting the manager's purchasing decisions
- Troubleshot front-desk computers and kitchen equipment on daily checks, reducing service disruptions

High Performance Computing Summer Institute, Jackson State University Jackson, MS | June 2024 - July 2024
Student Researcher

- Ran computational fluid dynamics simulations on a Linux HPC cluster, comparing flow across parameter settings
- Analyzed simulation output with Python and NumPy, visualizing velocity and pressure fields for team review
- Applied parallel computing methods from institute workshops, scaling runs beyond single-machine capacity

TECHNICAL PROJECTS

Options Heatmap Philadelphia, PA | December 2024 - Present
Sole Full-Stack Developer | Personal Project

- Developed a GEX/VEX heatmap for SPY, QQQ, IWM, and mega-cap tech, exposing key gamma levels at a glance
- Built a React/TypeScript UI and Node API fetching moomoo OpenD option chains, with automatic demo fallback
- Calculated GEX per 1% move from gamma, open interest, and spot price, netting call and put exposure by strike
- Estimated vanna exposure (VEX) via Black-Scholes when providers omit vanna, verified with Vitest unit tests

Prediction Stock Philadelphia, PA | August 2024 - Present
Sole Backend and Machine Learning Developer | Personal Project

- Designed and built a stock prediction platform end to end, using leakage audits to keep all features point-in-time
- Stored market data as immutable Parquet snapshots in a DuckDB catalog, making every experiment reproducible
- Ran 72 preregistered LightGBM and baseline trials on purged walk-forward folds, preventing look-ahead bias
- Shipped a read-only FastAPI service backed by 465 pytest tests, keeping every pipeline stage verifiable

Quant Model Philadelphia, PA | June 2022 - Present
Sole Full-Stack Developer | Personal Project

- Ingested 5 years of S&P 500, Nasdaq 100, CSI 300, and crypto data in Python, enabling cross-market research
- Implemented 10 trading signals, including RSI and MACD, with LightGBM ranking to surface top stocks
- Published backtests via PostgreSQL, Fastify APIs, and React dashboards, centralizing results in one view
- Automated refreshes every 4 hours for crypto and daily for equities via a worker, flagging stale data

HONORS, AWARDS, AND COURSEWORK

- 2nd Place, Ole Miss Hackathon (AI Course-Planning Advisor), University of Mississippi, April 2024
- 3rd Place, Ole Miss Hackathon Beta (C Spire Network Optimization), University of Mississippi, April 2023
- Dean's Honor Roll, University of Mississippi, Fall 2022

Relevant Coursework: Machine Learning, Database Systems, Data Structures and Algorithms, Software Engineering